

In this section we use non-linear least squares regression techniques to estimate the parameters of our model.

$$I = a_1 \cdot \text{Size}^{a_2} \cdot \sigma^{a_3} \quad (5.14)$$

$$MI = b_1 \cdot I \cdot \text{POV}^{a_4} + (1 - b_1) \cdot I \quad (5.15)$$

These parameters are:  $a_1, a_2, a_3, a_4$ , and  $b_1$ .

*Outliers.* To avoid potential issues resulting from outliers we filtered our data points based on daily stock volume and overall price movement. If a data point was outside of a specified range we excluded that data point. Filtering is commonly done on market impact data sets to avoid the effect of high price movement due to a force or market event that is not due to the buying or selling pressure of investors.

In our analysis we filtered the data to include only those data points with:

1. Daily Volume  $\leq 3 \cdot \text{ADV}$
2.  $\frac{-4 \cdot \sigma}{\sqrt{250}} \leq \text{Log Price Change (close-to-close)} \leq \frac{+4 \cdot \sigma}{\sqrt{250}}$

We decided to use four times the daily volatility to account for the potential incremental price movement due to the buying or selling pressure in an adverse momentum market. Analysts may choose to use different break points as well as filtering criteria.

*Factor Independence.* As is the case with any regression analysis we require explanatory factors to be independent. Unfortunately, our derivation process results in correlation across factors but this correlation is reduced by using multiple time horizons (full day, morning, and afternoon). Analysts can further reduce the correlation across factors through a sampling process of the data where we select a subset of data points such that the cross factor correlation is within a specified level (e.g.,  $-0.10 \leq \rho \leq 0.10$ ). Our resulting data set had  $N = 180,000$  points and the resulting correlation matrix is shown in [Table 5.3](#).

| Table 5.3 Factor Correlation Matrix |       |            |       |
|-------------------------------------|-------|------------|-------|
|                                     | Size  | Volatility | POV   |
| Size                                | 1     | −0.05      | 0.08  |
| Volatility                          | −0.05 | 1          | −0.03 |
| POV                                 | 0.08  | −0.03      | 1     |